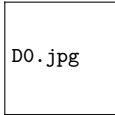


Database Management Systems

Lecture 11: Relational Algebra II — Joins and Expression Trees

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Outline

Natural Join: $R \bowtie S$

Natural Join $\bowtie$

$R \bowtie S$ is a join on all **common attributes** (same name and domain).

- 1 Compute $R \times S$
- 2 Select rows where common attributes are equal
- 3 Project out duplicate columns

SQL equivalent: NATURAL JOIN or explicit JOIN ... ON

Example:

Student(SID, Name, DeptID)

Dept(DeptID, DeptName)

Student

SID	Name	DeptID
S01	Amina	CS
S02	James	MA
S03	Priya	CS

Dept

DeptID	DeptName
CS	Computer Sci
MA	Mathematics

Result: Student $\bowtie$ Dept

SID	Name	DeptID	DeptName
S01	Amina	CS	Computer Sci
S02	James	MA	Mathematics
S03	Priya	CS	Computer Sci

Theta Join $\bowtie_{\theta}$ and Equijoin

Theta Join $\bowtie_{\theta}$

$$R \bowtie_{\theta} S = \sigma_{\theta}(R \times S)$$

A join with any **arbitrary predicate** θ (not just equality on common columns).

The predicate θ can use $=, \neq, <, \leq, >, \geq$ and AND/OR/NOT.

Equijoin

A theta join where θ uses **only equality** conditions.

Output retains both join columns (unlike natural join which eliminates duplicates).

Example — find students whose GPA exceeds their department average:

$\text{Student} \bowtie_{\text{Student.GPA} > \text{DeptAvg.AvgGPA}} \text{DeptAvg}$

Non-equijoin example — pairs of cities within 500 km:

$\text{City} \bowtie_{\text{distance} < 500} \rho_{\text{City2}}(\text{City})$

Natural Join Danger

Natural join joins on ALL common attribute names.

If two tables both have a column named *name* but they mean different things, natural join silently gives wrong results.

Outer Joins: Preserving Unmatched Tuples

Problem with Inner Join

A natural join (inner join) **drops** tuples that have no matching partner.

What if a student has no enrolled courses? That student disappears from the result.

Outer join variants:

- ▶ **Left outer** : keep all left tuples (fill right with NULLs)
- ▶ **Right outer** : keep all right tuples
- ▶ **Full outer** : keep all tuples

Student

SID	Name
S01	Amina
S02	James
S03	Priya
S04	Omar

Enrolment

SID	CourseID
S01	C101
S01	C102
S03	C101

Left Outer Join: Student Enrolment

SID	Name	CourseID
S01	Amina	C101
S01	Amina	C102
S02	James	NULL
S03	Priya	C101
S04	Omar	NULL

James and Omar have no

enrolments — preserved with NULL.

Expression Trees: Visualising Query Plans

Expression Tree

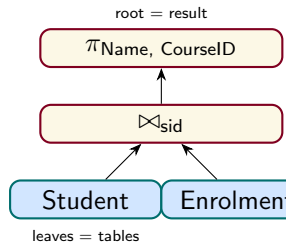
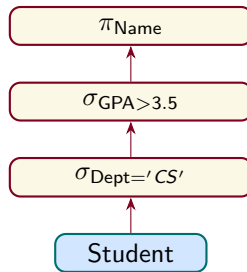
A **relational algebra expression tree** shows operations as nodes and relations as leaves.

- ▶ Leaves = base relations (tables)
- ▶ Internal nodes = operations (σ , π , $\bowtie$, ...)
- ▶ Data flows **bottom-up**
- ▶ Root = final result

The query optimizer transforms this tree to find the cheapest equivalent.

Query: Names of CS students with GPA > 3.5

$$\pi_{\text{Name}}(\sigma_{\text{GPA} > 3.5}(\sigma_{\text{Dept} = \text{'CS'}}(\text{Student})))$$



Key Takeaways

- 1 Natural join $\bowtie$: joins on all common attributes; eliminates duplicates
- 2 Theta join $\bowtie_{\theta}$: join with any predicate
- 3 Inner join = theta join; drops unmatched tuples
- 4 Outer joins $(, ,)$: preserve unmatched tuples with NULL
- 5 Expression trees visualise query evaluation order; basis of query optimisation

Next Lecture (L12)

Relational Algebra III — Division and Equivalence

- Division operator $\div$ (for “for all” queries)
- Query equivalence rules
- ICA: write algebra for 5 real queries

Reading

Silberschatz, **Ch. 6**: Formal Relational Query Languages (Sections 6.1–6.3)